



Course Outline Form
Department of Electrical and Electronic
Engineering School of Engineering
Brac University

EEE369/ECE369 Professional Practice, Engineers and Society

A. Course General Information:

Course Code :	EEE369/ ECE369
Course Title :	Professional Practice, Engineers and Society
Credit Hours :	3
Contact Hours/Week :	3
Category	School/Program Core
Type (Mandatory/Optional, Lecture/Laboratory/Project...):	Mandatory, Required, Engineering, Lecture
Prerequisites :	ENG 102 English Composition HUM 103 Ethics and Culture
Co-requisites:	None

B. Course Catalog Description (Content):

This course is designed to introduce undergraduate engineering students to the concepts, theory and practice of engineering professional ethics in the global and social context of contemporary engineering practices. This course will help students to explore what engineers do, to understand the social, political, legal and economic responsibility and accountability of the engineering profession as well as how engineering practice plays vital role in the development of sustainable growth, It will also allow students how to apply classical moral theory and take informed ethical decisions in engineering issues encountered in professional careers. The assessment of this course will be based on case-study and assignment based reports, presentations etc.

Introduction to Ethics and Professional Practice: Introduction, Ethics and professionalism, Scope of Ethics, Code of Ethics for Engineers,

Moral Reasoning and Ethical Theories-Ethical Theories, Duty Ethics, Right Ethics, Virtue Ethics, Utilitarianism, Self-Realization,

Codes of Ethics of Professional Engineering Societies and Reputed Engineering Companies-National and International- (National and International), THE INSTITUTION OF ENGINEERS, BANGLADESH (IEB) codes of ethics and rules of ethics, NSPE, IEEE, ASCE, ASME, ACM and Computer Ethics by Computer Institute, USA,

Ethics and Engineering Design, Understanding Ethical Problems and Solving Techniques- Line drawing, flow chart.

The Engineers Responsibility to Assess and Manage Risk and Ensure Safety- How Engineers Impose and Manage Risks, Sources of Risks, Engineering Approach to Defining Acceptable Risk, Risk Assessment Techniques, Communicating Risk and Public Policy.

Workplace Cultures, Responsibilities, Conditions, Labor Laws and Rights-Professional Duties, Professional Responsibilities and Professional obligation of Engineers Corrupt practices, Corruption & Bribery, Negligence, Professional Misconduct and Conflict of Interest, Workplace Responsibilities, Labor Law and Rights & Whistle blowing,

KAIZEN (5-S Techniques) for Improved Work Environment and Green Productivity, Ethics in Research-Honesty, Academic and Research Integrity, Research Misconduct, Responsible Authorship and Credit in Engineering Research,

Rights and Responsibilities Regarding Intellectual Property- Copyrights, "Fair Use," and the DMCA, Patents, Reverse Engineering

Environmental Ethics, Sustainability-Environmental Law and Policy - National and International, Life Cycle Analysis, Sustainable Development and Engineering Practice.

C. Rationale of the Course:

Introduction to professional practice of ethics and interaction of engineers with society. Ethics and professionalism, Ethical Terms, Source of Ethics, History of Codes & Engineering Ethics, Professional Duties, Professional Responsibilities and Professional obligation of Engineers Corrupt practices, Corruption & Bribery. Moral Frameworks-Ethical Theories, Duty Ethics, Right Ethics, Virtue Ethics, Utilitarianism, Self-Realization, Code of Ethics (National and International), Engineering / Professional Practice, Ethical Duties, Responsibilities and Obligations, Negligence, Professional Misconduct and Conflict of Interest. THE INSTITUTION OF ENGINEERS, BANGLADESH (IEB) codes of ethics and rules of ethics, NSPE, IEEE, ASCE, ASME, ACM and Computer Ethics by Computer Institute, USA. Honesty, Academic and Research Integrity, Ethical Dilemma and Moral Reasoning, ENGINEERING ETHICS FOR EXPERIMENTS and Learning from Past, Upholding and Commitment to Safety and Health, Workplace Responsibilities and Rights & Whistle blowing, and Environmental Ethics and Sustainability.

D. Course Objective

The objectives of this course are to

- Enable student to understand their role and responsibilities as engineering professionals through gaining knowledge of moral values, philosophies, professional code of ethics and practices
- Prepare students to be able to take informed ethical decisions when confronted with problems in the working environment
- Develop students' ability to assess the impact of engineering solutions in the broader societal and environmental context
- Enable students to evaluate the sustainability of engineering solutions
- Improve students' communication skills in regard to ethical and professional issues in engineering practice

E. Course Outcomes (Cos):

Upon successful completion of this course, students will be able to

Sl.	Course Outcome
CO1	Identify and value the responsibility of the engineers in regard to social, cultural, economic, legal, health, safety and welfare relevant to electrical and electronic engineering solutions and practice.

CO2	Evaluate the sustainability and impact of the electrical and electronic solutions in the broader societal and environmental context
CO3	Resolve competing and complex ethical issues related to the electrical and electronic engineering solutions and professional practices
CO4	Communicate effectively with regard to ethical, professional, societal and environmental issues in electrical and electronic engineering practices and solutions.

F. Mapping of Course Outcomes (COs) with Program Outcomes (POs):

COs	PO-a	PO-b	PO-c	PO-d	PO-e	PO-f	PO-g	PO-h	PO-i	PO-j	PO-k	PO-l
CO1						x						
CO2							x					
CO3								x				
CO4										x		

G. Mapping of CO-PO-Taxonomy Domain & Level-Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO1	Identify and value the responsibility of the engineers in regard to social, cultural, economic, legal health safety and welfare relevant to electrical and electronic engineering solutions and practice	f	Cognitive/ Analyze, Affective/ Valuing	Lecture notes, classroom discussion	Assignment, Case study report and/ or presentation
CO2	Evaluate the sustainability and impact of the electrical and electronic engineering solutions in the broader societal and environmental context	g	Cognitive/ Evaluate	Lecture notes, classroom discussion	Assignment, Case study report and/ or presentation
CO3	Resolve competing and complex ethical issues related to the electrical and electronic engineering solutions and professional practice	h	Affective/ Valuing	Lecture notes, classroom discussion	Assignment, Case study report and/ or presentation
CO4	Communicate effectively with regard to ethical, professional, societal and environmental issues in electrical and electronic engineering practices and solutions.	j	Affective/ Valuing Psychomotor/ Precision	Lecture notes, classroom discussion	Written report, Oral presentation

Mapping of CO - PO Indicators - Knowledge Profile

CO No.	PO No.	Applicable PO Indicator	Knowledge Profile
CO1	f	f1, f2	K7

CO2	g	g1,g2,g3	K7
CO3	h	h1,h2,h3	K7
CO4	J	j 1,j2,j4	

Description of PO Indicators

Program Outcome	Performance Criteria
(f) The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7)	f1. Demonstrate the contextual knowledge of assessing the societal, health, safety, legal and cultural issues in engineering solutions and professional practice.
	f2. Value the consideration of technical and non-technical issues related to societal, health, safety, legal and culture in developing the solution of complex engineering problem
(g) Environment and Sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex electrical and electronic engineering problems in societal and environmental contexts. (K7).	g1. Understand and evaluate the impact of engineering solutions to the environment and society
	g2. Understand the need for sustainable solutions of an engineering problem
	g3. Validate the sustainability of an engineering solution
(h) Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of electrical and electronic engineering practice. (K7).	h1. Maintain ethical standards in writing reports, preparing assignments, homework, exams etc.
	h2. Recognize and respect the professional ethics and responsibilities related to engineering practice
	h3. Apply the ethical principles related to the solution of an engineering problem
(j) Communication: Communicate effectively on complex activities with the electrical and electronic engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	j1. Write effective technical reports, prepare assignments, design documentation etc. on complex engineering activities
	j2. Prepare multi-media presentations, posters on engineering activities
	j4. Deliver effective oral presentations, participate in technical discussions on complex engineering activities

Description of Applicable Knowledge Profiles

Knowledge	Attribute	
K7	Comprehension	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability

H. Course plan specifying content, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs:

Class Number	Topic	Teaching-Learning Strategy	Assessment Strategy	Corresponding COs
I	ETHICS AND PROFESSIONAL PRACTICE			

1	Introduction, Engineering Professional Practice and Obligation, Engineers and society, Environment & sustainability and Ethics.	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Mid Exam	CO1, CO2
2	Ethical Terms, Source of Ethics, Brief History of Codes & Engineering Ethics, Ethics and Morality, Engineering Code of Ethics and professional Responsibilities for Engineers, Conflict of interest and Corrupt practices, Corruption & Bribery.	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Mid Exam	CO2, CO3
II	Moral Frameworks (Ethical Theories)			
3 & 4	Ethical Theories, Duty Ethics, Right Ethics, Virtue Ethics, Utilitarianism Ethics Self-Realization Ethics and Egoism.	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Mid Exam	CO1, CO2, CO3
III	Code of Ethics 1			
5	Importance of Code; Abuse of Codes; Limitations of Codes; Ethical Relativism; Justification of Codes; Engineering / Professional Practice, Ethical Duties, Responsibilities and Obligations, Negligence, Professional Misconduct and Conflict of Interest.	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Mid Exam	CO1, CO2, CO3
6&7	CODE OF ETHICS - THE INSTITUTION OF ENGINEERS, BANGLADESH (IEB), Rules of Ethics and Conduct for Engineers, The Code of Ethics for Procurement followed by Engineers in Bangladesh, International Code of Ethics for Engineering Profession	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Mid Exam	CO1, CO2

	(NSPE, IEEE, ASCE, ASME, ACM and Computer Ethics by Computer Institute, USA.			
IV	Honesty and Research Integrity			
8	Honesty, Truth and Truthfulness, AN AFRICAN FABLE, regarding battle of Truth and Falsehood and Trust & Trust worthiness, what is academic integrity? Research Integrity, Bias and Self-deception, Protecting Research Subjects, Giving and Claiming Credits, Survey, Survey Methods,	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Final Exam	CO2, CO3
V	Ethical Dilemma and Moral Reasoning			
9 & 10	Moral Choices and Ethical Dilemmas; Steps in Resolving Ethical Dilemmas; Right-Wrong or Better-Worse? Moral Decision Making as Design;	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Mid Exam	CO1, CO2
11	Assessment 1			CO1, CO2, CO3
12	Presentation 1			C04
VI	ENGINEERING ETHICS FOR EXPERIMENTS and Learning from Past			
15&16	Engineering as Experimentation, Engineers as Responsible Experimenters, Conscientiousness, Comprehensive Perspective – Relevant Factual Information, Moral Autonomy, Accountability, Informed Consent, A Balanced Outlook on Law, National and International Codes & Standards, Learning from the Past and challenges.	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Final Exam	CO1, CO2
VII	Upholding and Commitment to Safety and Health			
17,18 & 19	Safety and Risk, Concept of Safety, Concept of Risk, Acceptability of Risk, Assessing and Reducing Risk, Risk assessment method for assessing design safety, Design and development planning (ISO-9001) &	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Final Exam	CO2, CO3

	Practice in Safe Design, Uncertainties in Design, Testing for safety,			
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	Risk-Benefit Analysis, Personal Risk versus Public Risk, Examples of Improved Safety, Safe Exit, Managing risk and promoting safety are paramount in engineering practice, Taking responsibility for all health, safety and welfare related issues and code of practice for safe design of building & structure and engineering project/activities etc, Developing, implementing and improving appropriate hazard identification and risk management system and safety Complying ISO Standards, Law and Acts, Policy and Commitment for safe design, mitigation of risk and ensuring safety, health & no harm to environment.	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Final Exam	CO2, CO3
VIII	Workplace Responsibilities and Rights & Whistle blowing			
20 & 21	Work responsibility, Types of Workplace Responsibility, Employee Responsibilities in the Workplace, Responsibilities of engineer as employee and Leader of his team Responsibilities of Team Work & Managing Conflicts, Workplace Right, Confidentiality, Confidentiality and Management Policies, Conflicts of Interest, Loyalty and Collegiality, Professional Rights, Right of Professional Conscience, Refusal, Recognition, Privacy, Equal Opportunity – Non-discrimination, Employer Responsibilities on workplace safety, Whistle-Blowing, Types of Whistleblowing, Moral Guidelines, Protecting Whistle-blowers, Common Sense Procedures, Beyond Whistle-blowing.	Lecture, Notes, Lecture sheet, Team/ Group Work and Discussion	Quiz, Assignment, Case study, report and/or Presentation, Final Exam	CO2, CO3
IX	Environmental Ethics and Sustainability			
22,23 & 24	Fundamentals of Environment & Environmental Protection, Seven Environmental Principles, Environmental Issues. Environmental	Lecture, Notes, Lecture sheet, Team/ Group Work	Quiz, Assignment, Case study,	CO2, CO3

		and Discussion	report and/or Presentation,	
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	Impact Assessment (EIA), Environmental Ethics, Frameworks of Environmental Ethics and UN Conference on Human Environment in 1972 with Common Principles, Engineering Code References to the Environment & Sustainability and UN Conference on Environment & Development (Earth Summit) in 1992 with Common Principles. Commitment to keep environment clean, pollution free and sustainable and also use resources efficiently and effectively to ensure a sustainable environment and reduce and avoid the tragedy of the Commons. Undertaking engineering activities that contribute to sustainable development and responsibility for environmental, health, safety and welfare issues considering diverse impacts on technical, environment, social, cultural, economic, financial and global responsibility achieving sustainable development goal.		Final Exam	
25	Assessment			C01, C02, C03
26	Presentation 2			C04

I. Learning Materials:

Recommended Readings

- Ethics in Engineering by Mike W. Martin & Roland Schinzinger, 4th Edition, McGraw-Hill.
- Engineering Ethics Concepts and Cases by Charles E. Harris, Michael S. Pritchard & Michael J. Rabins, 4th ed, Wadsworth Cengage learning.
- Ethics in Engineering Practice and Research by Caroline Whitbeck 2nd ed, Cambridge university press.

Supplementary Readings

- Mike W. Martin, Roland Schinzinger, "Introduction to engineering ethics" 2nd ed, McGraw-Hill.
- Canadian Professional Engineering and Geoscience Practice and Ethics by Gordon C. Andrews 6th ed, Nelson Education
- International Standards for Risk, Safety, Health, Environment (ISO 9001-Risk Management, ISO 14001-Environment Management and ISO 45001-Occupational Health and Safety Management System)

J. Assessment and Evaluation:

Assessment Strategy

During the course duration, marking policy and guidance for students are mentioned as follows. These policies will be followed strictly and can only be amended as per the course instructor only.

- To assess the understanding and knowledge of the students, a total of 4 assessments will be taken, spread out evenly over the semester. Best 3 out of 4 will be counted.
- One mid-term and one final presentation will be taken during exam time. Students will be assessed on their level of attainment of CO4 through these presentations.
- Students will be given, case studies and comprehensive mini research topics on Ethical Issues in professional practice in order to assess their ability in understanding of ethical issues, and moral dilemma and also communications skills in regard to ethical and professional issues in engineering practice that meets the required specifications under certain constraints (CO2) (CO3) and (CO4).
- At least two Assignments will be given to assess CO1.

CO Assessment Plan:

Assessment Tools	Course Outcomes			
	CO1	CO2	CO3	CO4
Assignment	✓			
Case Study Report		✓	✓	✓
Presentation		✓	✓	✓

Marks distribution:

Assessment Tools	Weightage (%)
Class Performance & Attendance (70%)	10
Assignments (2 Nos)	20
Case Study Reports/Quizzes (3 Nos)	20
Mid Exam & Presentation (1)	25
Final Exam & Presentation (1)	25
Total	100

Make-up Procedures:

Students are not encouraged to give make-up mid-term except on certain special circumstances.

Make-up examinations will only be allowed to the students with extreme medical condition OR death in the immediate family during the semester. (Recommendations from 30th Syndicate and 69th Academic Council, Boot approved in the 27th meeting)

Student must notify in writing before the exam if he/she is going to miss the exam and needs to sit for a make-up exam. Students need to apply to sit for the make-up exam by filling out the appropriate form available at the office, getting the necessary approval and submitting the form to the registrar office with required fees for the make-up exam.

K. CO Attainment Policy:

As per the course outcome attainment policy of the Department of Electrical and Electronic Engineering

L. Grading policy:

As per Brac University grading policy

M. Course Coordinator:

Dr. A. S. Nazmul Huda

N. Course Teacher:

Mohaimenul Islam